Answer key: Lesson 10, Story Steps: Read It Twice

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Read it twice. Ben has 30 cards. He gives away 20. Then he gets 6 more.

Answer: Step 1: 30 - 20 = 10. Step 2: 10 + 6 = 16. Ben has 16 cards.

First, Ben gives away 20, so we take away: 30 - 20 = 10. Then he gets 6 more, so we add: 10 + 6 = 16. Ben has 16 cards now.

Watch for: Adding the 20 (30 + 20) because 20 is in the story.; Using 30 again in Step 2 instead of the 10 left after Step 1.

S2. Mei has 42 stickers. She gets 7 more. Then she gives 10 to Ali. Write a number sentence for each step.

Answer: Step 1: 42 + 7 = 49. Step 2: 49 - 10 = 39. Mei has 39 stickers.

Step 1: she gets 7 more, so add: 42 + 7 = 49. Step 2: she gives 10 away, so take away: 49 - 10 = 39. Mei has 39 stickers now.

Watch for: Using 42 again in Step 2 (42 - 10 = 32).; Adding both numbers (42 + 7 + 10).

S3. Ravi has 25 pencils. He buys 10 more. Then he loses 3. Circle the two number sentences Ravi needs. Then answer.

Answer: Circle 25 + 10 = 35 and 35 - 3 = 32. Ravi has 32 pencils.

Step 1: buys 10 more, so 25 + 10 = 35. Step 2: loses 3, so 35 - 3 = 32. The sentences 25 - 10 = 15 and 35 + 3 = 38 use the wrong sign.

Watch for: Circling 25 - 3 = 22 because both numbers are in the story.; Circling a sentence with the wrong sign.

S4. Story words tell us what to do. Write + for add or - for take away.

Answer: +, -, -, +

Finds more means add (+). Eats means take away (-). Gives away means take away (-). Join means more, so add (+).

Watch for: Choosing + for gives away because there is a number in the sentence.; Choosing - for join.

S5. Ali has 23 marbles. Ben has 10 more marbles than Ali. How many marbles do they have altogether? Draw if you like.

Answer: Step 1: Ben has 23 + 10 = 33. Step 2: 23 + 33 = 56. They have 56 marbles.

Step 1: Ben has 10 more than Ali, so 23 + 10 = 33. Step 2: altogether means add Ali and Ben: 23 + 33 = 56 marbles.

Watch for: Answering 33 (stopping after Step 1).; Adding 23 + 10 and calling that altogether.

S6. A bus has 30 children. 10 get off. Then 4 get on. Siti writes 30 + 10 = 40 and 40 + 4 = 44. What did Siti do wrong? Write the right steps and the answer.

Answer: Get off means take away: 30 - 10 = 20. Then 20 + 4 = 24. There are 24 children.

Children who get off leave the bus, so take away: 30 - 10 = 20. Then 4 get on, so add: 20 + 4 = 24. Siti added the 10 instead of taking it away.

Watch for: Not saying what went wrong (any clear sentence about get off meaning take away is fine).; Writing 44 again.

Think harder (page 3)

T1. Ravi has 15 marbles. Ravi has 3 more marbles than Mei. Hint: who has more, Ravi or Mei?

Answer: Mei has 12 marbles. Together they have 27.

Ravi has 3 more than Mei, so Mei has 3 fewer than Ravi: 15 - 3 = 12. Together: 15 + 12 = 27 marbles.

Watch for: Adding 15 + 3 = 18 because the word more is there.; Answering 12 for the second part.

T2. Mei is in a line to buy lunch. There are 6 children in front of Mei. There are 4 children behind Mei. How many children are in the line?

Answer: 11 children (6 + 4 + 1, counting Mei).

The children in front are 6 and behind are 4: 6 + 4 = 10. Mei is in the line as well, so 10 + 1 = 11 children.

Watch for: Answering 10 and leaving Mei out.

T3. Ben has 12 marbles. Mei has 5 more marbles than Ben. Siti has 3 fewer marbles than Mei. How many marbles has Siti?

Answer: Mei has 17. Siti has 14 marbles.

Step 1: Mei has 5 more than Ben: 12 + 5 = 17. Step 2: Siti has 3 fewer than Mei: 17 - 3 = 14 marbles.

Watch for: Taking 3 from 12 (Siti is compared with Mei, not Ben).; Answering 17 (stopping after Step 1).

Go further (page 4)

G1. A shop has 60 pencils. It sells 24 on Monday and 15 on Tuesday. On Wednesday a box of 30 pencils comes in. How many pencils does the shop have now?

Answer: Step 1: 60 - 24 = 36. Step 2: 36 - 15 = 21. Step 3: 21 + 30 = 51. The shop has 51 pencils.

Monday: 60 - 24 = 36. Tuesday: 36 - 15 = 21. Wednesday a box comes in, so add: 21 + 30 = 51 pencils.

Watch for: Adding 24 and 15 and taking that from 60 but forgetting the box.; Starting each step from 60 again.

G2. Here are two steps. Step 1: 25 + 10 = 35. Step 2: 35 - 8 = 27. Write a story that fits both steps. End with a question. Swap with a grown-up: do they get 27?

Answer: Any story where 25 grows by 10 and then 8 are taken away, and the question asks how many now. The answer is 27.

For example: Ali has 25 stickers. He gets 10 more. Then he gives 8 to his sister. How many stickers has Ali now? 25 + 10 = 35, 35 - 8 = 27.

Watch for: A story where the second step starts from 25 instead of 35.; No question at the end.

G3. Mei, Ali and Ben have 70 shells altogether. Mei has 25 shells. Ali has 10 more shells than Mei. How many shells has Ben?

Answer: Ali has 35. Mei and Ali have 60. Ben has 10 shells.

Step 1: Ali has 25 + 10 = 35. Step 2: Mei and Ali have 25 + 35 = 60. Step 3: Ben has the rest: 70 - 60 = 10 shells.

Watch for: Answering 35 or 60 (stopping early).; Taking only Mei's shells from 70.

Investigation (page 5): Does the order matter?

Both orders give the same answer (20 + 10 - 5 = 25 and 20 - 5 + 10 = 25) whenever both orders can be done. But the order can matter for what you are allowed to do: from 4 you cannot take away 6 first, though 4 + 10 - 6 = 8 works. So: the answer is the same, but you cannot take away more than you have.

Look for: A child who tests several numbers, says what they notice, and finds a case where taking away first is not possible (start 4, take away 6).

Exit ticket (page 5)

E1. A bus has 25 children. 10 get off. Then 3 get on. How many children are on the bus now?

Answer: 18 (25 - 10 = 15, then 15 + 3 = 18)

Get off means take away: 25 - 10 = 15. Get on means add: 15 + 3 = 18.

Watch for: Writing 38 (adding the 10).; Writing 15 (stopping after Step 1).

E2. Ben has 6 stickers. Mei has 5 more stickers than Ben. How many stickers has Mei?

Answer: 11 (6 + 5)

Mei has more than Ben, so add: 6 + 5 = 11.

Watch for: Writing 1 (taking away).; Writing 5.

E3. Ali has 7 fewer marbles than Mei. Mei has 20 marbles. How many marbles has Ali?

Answer: 13 (20 - 7)

Ali has fewer, so take away: 20 - 7 = 13.

Watch for: Writing 27 (adding because the number 7 is there).; Writing 7.

Sneak-peek puzzles (not in the workbook)

1. 7 stars.
2. 11. First find Ben's marbles: 4 + 3 = 7. Then add Ali's 4: 4 + 7 = 11.